

Developmental epidemiological courses leading to antisocial personality disorder and violent and criminal behavior: Effects by young adulthood of a universal preventive intervention in first- and second-grade classrooms

Hanno Petras^{a,e,*}, Sheppard G. Kellam^{b,e}, C. Hendricks Brown^c, Bengt O. Muthén^d,
Nicholas S. Ialongo^e, Jeanne M. Poduska^b

^a University of Maryland College Park, Department of Criminology and Criminal Justice, College Park, MD 20742, USA

^b American Institutes for Research, Baltimore, MD 21230, USA

^c Department of Epidemiology and Biostatistics, College of Public Health, University of South Florida, Tampa, FL 33612, USA

^d Graduate School of Education and Information Studies, UCLA, Los Angeles, CA 90095-1521, USA

^e Johns Hopkins University, Bloomberg School of Public Health, Baltimore, MD 21205, USA

Received 5 April 2007; received in revised form 7 September 2007; accepted 19 October 2007

Available online 19 February 2008

Abstract

Background: Antisocial personality disorder (ASPD), violent and criminal behavior, and drug abuse disorders share the common antecedent of early aggressive, disruptive behavior. In the 1985–1986 school year teachers implemented the Good Behavior Game (GBG), a classroom behavior management strategy targeting aggressive, disruptive behavior and socializing children to the student role. From first through seventh grade the developmental trajectories of 2311 students from 19 Baltimore City Public Schools were examined. We report the GBG impact on these trajectories and ASPD and violent and criminal behavior by age 19–21.

Methods: In five urban, poor to lower middle class predominately African-American areas, three to four schools were matched and within each set randomly assigned to one of three conditions: (1) GBG, (2) a reading achievement program, or (3) the standard program. Classrooms and teachers were randomly assigned to intervention or control. Measures at 19–21 included self reports and juvenile court and adult incarceration records. GBG impact was assessed via General Growth Mixture Modeling based on repeated measures of aggressive, disruptive behavior.

Results: Three trajectories of aggressive, disruptive behavior were identified. By young adulthood, GBG significantly reduced the rates of ASPD and violent and criminal behavior among males in the *persistent high* aggressive, disruptive trajectory.

Replication: A replication was implemented with the following cohort of first-grade children using the same teachers, but with diminished mentoring and monitoring. Beneficial impact was found among *persistent high* males through seventh grade. By young adulthood GBG effects on ASPD and violent and criminal behavior were non-significant, but generally in the hypothesized direction.

© 2007 Elsevier Ireland Ltd. All rights reserved.

Keywords: Good Behavior Game; Developmental epidemiology; Growth mixture modeling; Universal prevention; Classroom behavior management; Aggressive, disruptive behavior; Antisocial personality disorder; Violence and criminal behavior

1. Introduction

Antisocial personality disorder (ASPD) is one of the more common serious mental health problems presently faced in the U.S. (Kessler et al., 1994; Moran, 1999; Turner and Gil,

2002). It is associated with both economic costs and great suffering for individuals (e.g., involvement with the legal system, co-occurring substance use and other mental health problems) families (e.g., broken homes, non-support), the community (e.g., victims, crime and aggression) and the state (e.g., welfare, imprisonment) (Britt, 2000; Capaldi and Stoolmiller, 1999; Cohen and Miller, 1998; Krueger et al., 1998; Lambert et al., 2001; Loeber et al., 2000; New and Berliner, 2000; Robinson and Keithley, 2000). ASPD is frequently comorbid with vir-

* Corresponding author. Tel.: +1 301 405 4716.

E-mail address: hpetras@crim.umd.edu (H. Petras).

tually every drug abuse disorder (Compton et al., 2005), and it increases the morbidity and complicates the treatment for substance use disorders (Westermeyer and Thuras, 2005). Both ASPD and drug abuse disorders are also commonly associated with serious violent and criminal behavior. Importantly, heavy use of alcohol and other drugs appears to increase risk for both commission and victimization of violent acts (Chaiken and Chaiken, 1990). In addition, Fazel and Danesh (2002) report in a review of serious mental disorders in special populations that close to half (47%) of imprisoned individuals qualify for an antisocial personality disorder diagnosis. Importantly, all of the above-mentioned outcomes associated with ASPD, drug abuse, and violence require high levels of service use (Poduska et al., 2008; Webster-Stratton and Taylor, 2001).

Given these adverse effects and the difficulty in treating ASPD and its co-occurring antisocial outcomes once they are established (Reid and Thorne, 2006), there has been a growing consensus among social scientists and policy makers that early preventive interventions targeting the antecedents of antisocial behavior are preferable to treatment approaches (Bennett et al., 1998, 1999; Mrazek and Haggerty, 1994; Offord and Bennett, 1997; Reiss and Price, 1996). In light of this strong evidence regarding the continuity between childhood and adult antisocial behavior, this preventive approach is especially pertinent (Farrington, 1995; Robins, 1966). Indeed, decades of research suggest that prevention is the most effective strategy available for reducing youth antisocial and violent behavior (Dodge, 1999; Hawkins et al., 2000b; Satcher, 2001).

Early aggressive, disruptive behavior is widely recognized as an antecedent of later ASPD and violent and criminal behavior as well as drug abuse and school dropout (Dishion et al., 1996; Ensminger et al., 1983; Ensminger and Slusarcick, 1992; Farrington and Gunn, 1985; Hawkins et al., 2000a; Kellam et al., 1983, 1991, 1994, 2008; Kershaw, 1992; McCord and Ensminger, 1997; Robins, 1978; Sameroff, 1994; Schwartzman et al., 1985). Given the importance of early aggressive, disruptive behavior as a precursor for an array of later problems, a number of universal (i.e., directed at the entire population) classroom preventive intervention trials have targeted this antecedent as a means of preventing later antisocial behavior and drug abuse. Examples of rigorous trials incorporating random assignment include The Seattle Social Development Project (Hawkins and Weiss, 1985), Linking the Interests of Families and Teachers (Eddy et al., 2000; Reid et al., 1999), Promoting Alternative Thinking Strategies (Greenberg and Kusche, 1998), Second Step (Grossman et al., 1997), Bullying Prevention (Olweus, 1991, 1992; Olweus and Alsaker, 1991; Smith and Sharpe, 1994), and the three generations of developmental epidemiology-based field trials involving the Good Behavior Game (GBG) in the Baltimore City Public School System (BCPSS) (Kellam et al., 2008). These randomized trials are particularly important and powerful in determining the efficacy of interventions such as the GBG and in testing the etiologic model upon which the GBG is based (Brown and Liao, 1999; Brown et al., 2008; Farrington and Welsh, 2005; Kellam and Langevin, 2003; Kellam and Rebok, 1992; Kellam et al., 1999).

The Baltimore trials have a study design with two unique features that make it well suited for the study of long-term outcomes. First, study participants were followed into young adulthood, which gives these trials the capacity to inform the field about the long-term impact of the GBG on a range of outcomes including antisocial and violent and criminal behavior as the subjects move into adulthood. Although a number of the intervention programs targeting early aggressive, disruptive behavior cited above have demonstrated short-term impact, the Baltimore prevention intervention research is part of only a small set of programs that followed study participants into the young adult years (for a comprehensive review of school-based violence prevention programs, see Gottfredson, 2001; Gottfredson et al., 2002). Second, the Baltimore trials are part of only a few to employ a randomized block design to disentangle the effects of the school from the effects of the intervention itself (for more detail, see Kellam et al., 2008).

The GBG is a team-based behavior management strategy that promotes appropriate classroom behavior by rewarding teams of students that do not exceed maladaptive behavior standards (Barrish et al., 1969; Dolan et al., 1993; Kellam et al., 1994). The goal is to encourage students to manage their behavior through group reinforcement and mutual self-interest. Previous research has documented the effectiveness of the GBG in reducing the level and development of aggressive, disruptive behavior, particularly among aggressive, disruptive males (Brown, 1993; Dolan et al., 1993; Kellam and Anthony, 1998; Kellam et al., 1994, 1998; Muthén et al., 2002; Rebok et al., 1996).

1.1. The life course/social field theory and developmental epidemiology

Life course/social field theory has guided the use of preventive interventions aimed at early antecedent of later problem outcomes as well as the conceptualization of normal and pathologic development in the Baltimore trials (Kellam et al., 1975; Kellam and Rebok, 1992). Central to life course/social field theory is the concept that individuals face specific social task demands in various social fields over the major stages of the life course. The specific social task demands the individual confronts through each stage of life are defined by individuals in each social field, whom we have termed the *natural rater(s)*. The natural rater not only defines the tasks but also rates the adequacy of performance of the individual in that social field. Parents function as natural raters in the family, peers in the peer group, teachers in the classroom, and supervisors in the workplace. This interactive process of demand and response is termed *social adaptation*, and the judgment of adequacy of the individual's performance by natural raters is labeled *social adaptational status* (Kellam et al., 1975).

In line with the organizational approach to development (Cicchetti and Schneider-Rosen, 1984), normal development is viewed within the life course/social field theory as marked by the integration of earlier competencies into later modes of function, with the earlier competencies remaining accessible, ready to be activated and utilized during times of stress, crisis, novelty, and creativity. It follows then that early successful social adaptation

in the face of prominent developmental challenges tends to promote later adaptation as the individual traverses the life course and encounters new and different social task demands across the main social fields (Cicchetti and Schneider-Rosen, 1984).

A critical developmental challenge early in the life course is entrance into the classroom social field, where the child is confronted with the teacher's demands for academic achievement, compliance, attention, and participation in classroom and peer activities. The transition into middle school and early adolescence provides another set of developmental tasks and challenges, which center on separation and individuation from parents and the growing demands of the family, classroom, and peer group social fields. Life course/social field theory posits that success in meeting these evolving and new task demands in middle school will depend, in part, on successful adaptation to the earlier developmental challenges encountered during the transition to elementary school. This key developmental principle, supported by a growing empirical literature, forms the basis for the focus in the Baltimore trials on successful adaptation to first grade as a means of improving social adaptational status over the life course.

1.2. Developmental pathways of aggressive, disruptive behavior

Several researchers have worked on theoretical and empirical modeling of aggressive, disruptive behavior development (Loeber et al., 1998; Moffitt, 1993; Moffitt et al., 1996, 2002; Nagin and Tremblay, 1999; Nagin et al., 1995; Patterson et al., 1992, 1998). Despite differences in terminology and emphasis, each model identifies two to five distinct groups of antisocial youth with different behavioral patterns, risk factors, and prognoses for desistance from antisocial behavior as adults. Each model proposes one to two chronic groups whose early and persistent aggressive, disruptive behavior is likely to be related to a biological or genetic vulnerability that is exacerbated by poor parenting and early school failure (i.e., the *early starters*, Patterson, 1982; *life course persistent groups*, Loeber and Stouthamer-Loeber, 1998; Moffitt, 1993). Each model also identifies one to two less severe groups whose antisocial behavior starts later, is less aggressive, disruptive, more sporadic, and stems from later socialization experiences such as deviant peer affiliations in early adolescence (i.e., the *late starters*, Patterson et al., 1992; *adolescent-limited*, Moffitt, 1993; and *limited duration pathways*, Loeber and Stouthamer-Loeber, 1998). Also implicit in each model is the assumption that there is at least one other group of youth who do not exhibit problems with antisocial behaviors. These models have helped to shift the study of youth antisocial behavior away from a variable-centered focus on describing broad predictors of behavior toward a more person-centered focus emphasizing individual differences in development (Magnusson, 1998). However, the empirical literature has focused primarily on the development of male aggression and has given little attention to the role of early aggression and other disruptive behaviors in the development of later female antisocial behavior (Keenan and Shaw, 1997; Silverthorn and Frick, 1999). Although some empirical studies

examining pathways of antisocial behavior development among females are starting to emerge in the literature (e.g., Broidy et al., 2003; Cote et al., 2001; Schaeffer et al., 2006), research regarding gender differences in these pathways is limited.

1.3. Intervention theory

The mechanisms by which the GBG intervention is hypothesized to reduce the risk for later antisocial behavior is consistent with the integration of the life course/social field theory and the early starter and late starter models of the development of antisocial behavior identified by Patterson et al. (1992). The early starter models of antisocial behavior build upon the theory of coercive family process developed by Patterson (1982). In the toddler years, parents' failure to effectively punish noncompliant and aggressive, disruptive behavior comprises the first step in a process which serves to train the child to become progressively more coercive and antisocial. In the classroom setting, such children prove difficult for teachers, peers, or other natural raters to teach appropriate forms of social interaction and problem solving. Moreover, their coercive style may be further reinforced in the presence of inconsistent and punitive teacher disciplinary practices. Ultimately parents, teachers, and well-adjusted peers reject the coercive child, which results in the child's failure to develop academic, social, and occupational survival skills. Patterson et al. (1992) argues that a lack of adequate monitoring by parents in early adolescence and rejection by teachers and mainstream peers precipitates a child to drift into a deviant peer group, wherein a wide array of antisocial and delinquent behavior, including alcohol and drug use, may be reinforced, along with a rejection of mainstream norms and mores (Brook et al., 1989; Jessor and Jessor, 1978; Patterson et al., 1992).

A second commonly identified trajectory of aggressive, disruptive behavior involves late onset (i.e., in the pre- or early adolescent years) antisocial behavior and substance use. Patterson et al. (1992) argues that late starters typically exhibit marginal levels of social adaptation in the elementary school years in terms of aggressive, disruptive behavior, and their caregivers' discipline and monitoring skills may be marginal at best. Consequently, these children are quite vulnerable to disruptions in parental monitoring and supervision, which may lead to rapid escalation of behavior and/or academic achievement problems. More specifically, Patterson et al. (1992) hypothesizes that the escalation in antisocial behavior seen in these late starters in early adolescence is the product of perturbations in parental monitoring and supervision brought on by serious family adversities that first surface in the middle school years. The disruptors may include a divorce, serious financial distress associated with the loss of a job, and/or the late onset of parental psychiatric distress or substance use, abuse, and/or dependence. Like the early starters, late starters may be rejected by their mainstream natural raters, as a result of their coercive and antisocial behavior. Their limited social skills and the rejection by their mainstream natural raters can precipitate an individual to drift into a deviant peer group, where antisocial behavior, substance use, and rejection of mainstream social values, mores, and institutions are reinforced.

In keeping with life course/social field theory and its integration with Patterson et al.'s (1992) early starter developmental model, the Baltimore GBG intervention is hypothesized to reduce the early aggressive, disruptive behavior and its distal correlates by improving teachers' disciplinary practices. This improved classroom management will then result in a reduction of early aggressive, disruptive and coercive behavior at the individual and the classroom level, and as a result there will be fewer opportunities for youth to learn inappropriate behavior through modeling of their classmates' aggressive, disruptive behavior. The youth will then be at decreased risk of being rejected by parents/caregivers, teachers, peers, and other natural raters and thus less likely to drift into a deviant peer group, where antisocial behavior and substance use may be reinforced and mainstream norms and mores rejected. Consequently, the youth will then be at reduced risk of serious antisocial behavior, violent and criminal behavior, and drug abuse in adolescence and adulthood.

With respect to the late starter model and mechanisms of GBG intervention impact, it is hypothesized that youth in the GBG classrooms will maintain higher levels of social adaptation in the face of disruptions in parent supervision, discipline, and reinforcement in the pre- to early adolescent years than their standard setting counterparts. As a result, these youth will be less likely to drift into a deviant peer group and engage in serious antisocial behavior and substance abuse in adolescence and adulthood. Finally, those youth who continually exhibited minimal aggressive, disruptive behavior have shown successful social adaptation and therefore are hypothesized not to be influenced by other students' aggressive, disruptive behavior.

1.4. The present paper

In this paper, we report on the impact of the GBG on the course of aggressive, disruptive behavior from first to seventh grade and on the young adult outcomes of ASPD and violent and criminal behavior in young adulthood (age 19–21) for both males and females from the first generation of the Baltimore trials. This is an extension of previously published reports (Dolan et al., 1993; Kellam et al., 1994), which documented the more proximal or shorter term, impact of the intervention. In Kellam et al. (2008) the impact of the GBG is briefly reported, taking into account baseline aggressive, disruptive behavior in the fall of first grade. In this paper we examine the developmental trajectories and variation in impact of the GBG along with extending the young adult outcomes to include violent and criminal behavior. Importantly, we compare the results obtained using the data from the effectiveness trial (i.e., Cohort 1) to the data from the sustainability trial (i.e., Cohort 2).

In accord with the existing evidence regarding the development of aggressive, disruptive behavior, we hypothesize that there will be three developmental trajectories of aggressive, disruptive behavior with varying risk for later antisocial and violent and criminal behavior (Muthén et al., 2002; Petras et al., 2004; Schaeffer et al., 2003, 2006). We hypothesize that children, especially males who are highly aggressive, disruptive early in elementary school and who remain aggressive, disruptive over elementary and middle school years (i.e., a *persistent high* class),

will be at greater risk for later ASPD and violent and criminal behavior than their less aggressive, disruptive counterparts. We also hypothesize a second group of children who enter school with lower levels of aggressive, disruptive behavior and then escalate by the end of elementary school/early middle school to a higher level of aggressive, disruptive behavior (i.e., an *escalating medium* class). We hypothesize this group will also be at an elevated risk for later ASPD and violent and criminal behavior compared with their less aggressive, disruptive counterparts. Lastly, we hypothesize a third group of children who display a consistent low level of aggressive, disruptive behavior from first to seventh grade (i.e., a *stable low* group) and who will show the lowest level of risk for later ASPD and violent and criminal behavior. In addition, we hypothesize a lesser GBG impact for females given their overall lower levels of aggressive, disruptive behavior (Enslinger et al., 1983; Kellam et al., 1983; Schaeffer et al., 2006; Wilcox et al., 2008).

2. Methods

2.1. Study design

Study participants were first-grade students from 19 Baltimore City Public Schools who were initially assessed in the fall of first grade as part of a randomized field trial of two classroom-based interventions targeting either early learning with mastery learning (ML) or aggressive, disruptive behavior with the GBG. The schools were drawn from five geographic areas within the eastern half of the city that were defined by census tract data and vital statistics obtained from the Baltimore City Planning Office. The areas varied by ethnicity, type of housing, family structure, and income, unemployment, violent crime, suicide, and school drop out rates. Three or four schools were matched within each of the five areas.

Special education and gifted classrooms were excluded from the pool of potential classrooms in light of the fact that the preventive interventions targeted regular or mainstream classrooms. In schools with three or fewer regular first-grade classrooms, all classrooms participated, and in larger schools, three first-grade classrooms were randomly selected for inclusion in the study.

Of the three or four matched schools selected in each of the five urban areas, we randomly assigned schools to serve as (1) schools in which the GBG would be tested independently of ML, (2) schools in which ML would be tested independently of the GBG, or (3) schools that would serve as external control schools in which neither of these interventions would be implemented, rather the standard program would be in place. These external control schools offered protection against the possibility of between-classroom contagion that might occur between intervention classroom and internal control classrooms within the same school. Following the school assignment, children were assigned to classrooms in a manner designed to create comparability across classrooms within each school. Then, first-grade classrooms within the GBG and ML schools were randomly assigned to either the intervention or internal control condition. Internal control classrooms were used to hold constant school building, family, and/or community differences such as the effect of the principal on the school environment, for example. Internal control classrooms composed the control group for the present study (for more detail see Kellam et al., 2008).

2.2. The population samples

Two cohorts with a total of 2311 students were originally available within the 19 participating Baltimore City Public Schools in first grade from recruitment in the 1985–1986 and 1986–1987 school years. We describe the two populations together here but remind the reader that the effectiveness trial involved the first cohort, while the second cohort was a replication with less mentoring and monitoring, thus considered to be a sustainability trial (see Kellam et al., 2008). Of the population, 1531 (66.2%) were assigned to intervention conditions not pertinent to this paper (i.e., external control, internal ML control, and ML classrooms)

Table 1
Sample description

	Cohort 1 (N = 403)	Cohort 2 (N = 365)	Difference test
Male	199 (49.4%)	181 (49.6%)	n.s.
African-American	302 (74.9%)	266 (72.9%)	n.s.
Free/reduced-price lunch	209 (51.9%)	267 (73.2%)	$p < 0.001$
GBG intervention	238 (59.1%)	208 (57.0%)	n.s.
Fall of first-grade aggression	2.0 (S.D. = 1.1)	1.8 (S.D. = 0.9)	$p < 0.010$
Age in fall of first grade	6.4 (S.D. = 0.5)	6.3 (S.D. = 0.4)	$p < 0.011$
ASPD			
ASPD diagnosis prevalence	62 (20.2%)	62 (22.0%)	n.s.
Amount missing	96	83	
ASPD and violent and criminal behavior			
Both	27 (8.8%)	23 (8.2%)	n.s.
ASPD only	35 (11.4%)	39 (13.8%)	
Violent and criminal behavior only	24 (7.8%)	18 (6.4%)	
Neither outcome	221 (72.0%)	202 (71.6%)	
Amount missing	96	83	
Number of missing points			
0–1	(48.9%)	(29.9%)	$p < 0.001$
2–3	(17.6%)	(32.9%)	
4–5	(19.1%)	(14.5%)	
6–9	(14.4%)	(22.7%)	

and were not included in this analyses. In addition, twelve students did not have valid teacher ratings of aggressive, disruptive behavior and were consequently excluded from the analysis.

Of the remaining 768 participants, 403 (52.5%) were in the first cohort and 365 (47.5%) in the second cohort (see Table 1). Gender was equally distributed in both cohorts and the majority of the sample was African-American (see Table 1). There was a significant difference between the cohorts on participation in the free or reduced-price lunch program, a proxy for family income; the percentage of students in Cohort 1 receiving a free or reduced-price lunch was 51.9%, while in Cohort 2 it was 73.2% ($\chi^2 = 18.855$, d.f. = 1, $p < 0.001$). Upon entrance into first grade, youth in the first cohort ranged in age from 5.08 to 9.42 years with a mean age of 6.4 years, while youth in the second cohort were significantly younger with ages ranging from 5.56 to 7.93 years with a mean age of 6.3 ($p < 0.011$). Teacher ratings of aggressive, disruptive behavior in the fall of first grade were slightly lower for Cohort 2 than they were for Cohort 1 ($p < 0.01$). Overall, no significant cohort differences were found in respect to the students' ethnic background and intervention condition. For the age 19–21 young adult follow-up, 76.2% of the Cohort 1 and 77.3% of the Cohort 2 youth contributed to the data used in this report.

2.2.1. Assessment design. For Cohort 1, teacher reports of child aggressive, disruptive behavior were gathered twice a year in first and second grades, once a year during third through seventh grade, and at the age 19–21 follow-up assessments. The assessments in the fall of first grade were carried out prior to the initiation of the interventions. Data gathered in the first-grade assessments consisted of teacher reports of child aggressive, disruptive behavior and intervention assignment. At ages 19–21, a follow-up structured clinical interview was used to ascertain whether the participant met criteria for antisocial personality disorder. In addition, records of violent and criminal behavior were obtained at the time of the young adult follow-up interview and repeated yearly searches were conducted thereafter. The assessment design was identical for Cohort 2 except data was not gathered in the fall of second grade because of funding issues.

2.3. Measures

2.3.1. Aggressive, disruptive behavior. Teacher ratings of aggressive, disruptive behavior were obtained using the Teacher Observation of Classroom Adaptation-Revised (TOCA-R; Werthamer-Larsson et al., 1991). The TOCA-R

is a structured interview with the teacher administered by a trained assessor. Teachers respond to 36 items pertaining to each child's adaptation to classroom task demands over the last three weeks. The level of adaptation is rated by teachers on a six-point frequency scale (1 = almost never through 6 = almost always). We used the authority acceptance subscale which includes the following items: (1) breaks rules, (2) harms others and property, (3) breaks things, (4) takes others property, (5) fights, (6) lies, (7) trouble accepting authority, (8) yells at others, (9) stubborn, and (10) teases classmates. The internal consistency reliability coefficients for the aggressive, disruptive behaviors subscale ranged from 0.92 to 0.94 over grades one through seven, or ages eight to thirteen. The 1-year test–retest intraclass reliability coefficients ranged from 0.65 to 0.79 over grades two to three, three to four, four to five, five to six, and six to seven. Teacher reports of aggressive, disruptive behavior have also been found to have the highest level of agreement with students' self reports, and have been shown to be equivalent or superior in the level of agreement between parent and student self reports of problem behavior (Lochman, 1995; Loeber et al., 1984). Scores on the aggressive, disruptive behavior subscale were significantly related to the incidence of school suspensions within each year from grades one to seven, i.e., the higher the score of aggressive, disruptive behavior, the greater likelihood of being suspended from school that year.

2.3.2. Antisocial personality disorder diagnoses. As part of a larger follow-up telephone interview at age 19–21, a scale was developed and administered to determine whether the participant met Diagnostic and Statistical Manual-IV (American Psychiatric Association, 1994) criteria for antisocial personality disorder (ASPD) including a history of conduct disorder. The questions comprising the scale were keyed to DSM-IV criteria and the diagnoses derived in accord with those criteria (Turner and Gil, 2002). To reduce the likelihood of socially desirable responses, participants were asked to maintain their own count of "yes" responses as opposed to responding "yes" or "no" to the interviewer's questions, and to insure against respondents losing track of the count, they were asked to have a pencil and sheet of paper available to mark down the number of "yes" responses. In addition, the questions were divided into three sections and the count of "yes" responses was obtained by the interviewer at the end of each section.

2.3.3. Violent and criminal behavior. Juvenile court and adult incarceration records were obtained at the time of the young adult follow-up interview (i.e.,

between age 19 and 21) and repeated yearly searches were conducted thereafter. For this paper we use information from the search in February of 2006. A juvenile court record for a violent crime (e.g., assault, rape) was obtained from local records in Baltimore City. The use of the juvenile data was supported in writing by the Chief Judge of the Baltimore Juvenile Court and was included in the Hopkins JHSPH IRB.

Records of incarceration for an offense classified as a felony in the Uniform Crime Reports system (i.e., murder, non-negligent manslaughter, forcible rape, robbery, aggravated assault, burglary, larceny-theft, and motor vehicle theft) was used as an indicator of violent and criminal behavior as an adult. These data were obtained from the Maryland Department of Correction and are considered public record. The juvenile court records and adult incarceration records were used to characterize an individual as having a record of violent and criminal behavior.

By integrating information about the ASPD diagnosis as well as records of violent and criminal behavior, four categories were established: both an ASPD diagnosis and a record of violent and criminal behavior, only a diagnosis of ASPD, only a record of violent and criminal behavior, or neither outcome by young adulthood. Among the 269 males with data on this outcome, 15.6% had both outcomes, 17.5% met criteria for ASPD only, 11.2% had a record of violent and criminal behavior only, and the remaining 55.8% had neither outcome. Among females with data, 2.5% had both outcomes, 8.4% had a diagnosis of ASPD, 3.8% had a record of violent and criminal behavior only, and 85.3% had neither outcome.

In terms of construct validity, males with an ASPD diagnosis were three times more likely than those who did not to have a record of violent and criminal behavior (ASPD, odds ratio (OR): 3.40; 95% CI: 2.38–4.83). In comparison, females diagnosed with ASPD were four times more likely to have a record of violent and criminal behavior (OR = 4.08; 95% CI = 2.32/7.19).

2.4. Analytic plan

We used general growth mixture modeling (GGMM) as described in Muthén et al. (2002) and Muthén (2004) to identify distinct developmental trajectories of aggressive, disruptive behavior. GGMM clusters growth curves into a small number of classes, each of which can be examined independently for intervention impact. GGMM additionally offers the opportunity to examine how the probability of a young adult outcome (e.g., ASPD and violent and criminal behavior) depends on class and intervention. Each class of curves in GGMM provides an interpretable pattern of growth based on its mean growth curve.

We first fit growth mixture models to the nine time points (eight time points in Cohort 2) at which teachers rated students' level of aggressive, disruptive behavior, refining the fit using Bayesian Information Criterion (BIC) to compare alternative models. We then evaluated the relationship between class membership and the young adult outcomes as well as the impact of the intervention on the class specific slopes and on the outcomes in young adulthood for females and males in Cohort 1 separately. The analysis was replicated for Cohort 2. In the models we report log likelihoods and numbers of parameters. Because these models fit both continuous and discrete data, the log likelihoods themselves are not meaningful, but differences in log likelihoods from nested models (i.e., likelihood ratio test) do result in appropriate χ^2 tests, which we report as χ^2 along with their degrees of freedom. We also tested for the class-specific impact of the GBG using Wald-type tests that compared the appropriate coefficient to its standard error. In the few cases where these tests functioned poorly – for example, a log odds ratio with a cell probability of zero – we used the above-mentioned likelihood ratio tests. Testing classes for intervention impact is similar to the process of examining mediation (Baron and Kenny, 1986); however, a straightforward test for mediation in the growth mixture modeling framework is not readily available nor is the procedure appropriately documented in the published literature. The goal in growth mixture modeling is to derive classes with limited within-class variance in terms of the growth parameters (intercept and slope); therefore, if one wished to test for mediation using the within-class slope as a mediator, intervention condition as the independent variable, and antisocial personality disorder as the outcome, for example, slope variance would be too small to provide an adequately powered test of mediation. In the absence of an established method to test for mediation, we use a parallel procedure; the young adult outcome in growth mixture modeling is regressed on the categorical vari-

able representing class membership and not the within-class growth parameters (intercept and slope). The use of the likelihood ratio test to compare competing models can be best described as having comparability to the Baron and Kenny (1986) method.

The impact of the GBG intervention through seventh grade on growth of aggressive, disruptive behavior was assessed by examining the effect of the GBG intervention on the linear and nonlinear slope parameters within each of the trajectory classes as well as conducting an overall likelihood ratio test on two degrees of freedom. For the models we have tested here, these two approaches all lead to the same conclusions of significant intervention effect over time, so we only present results using the former approach. Given a significant intervention impact in a particular class we further investigated the length of time for which this effect was sustained. For this purpose, we constructed a test statistic which related the time specific estimated mean differences between the two conditions (i.e., GBG vs. control) and to the time specific variance and covariance components. This statistic approximates a z -distribution, and the corresponding p -value can then be computed to test the hypothesis that the two means are not significantly different from zero. We report these results where significant from a univariate perspective. We also report those grade levels where the GBG impact reached significance after controlling for the multiple testing through the use of Bonferroni's or Scheffé's method, which produce virtually identical confidence intervals in this particular case. The added conservativeness of these procedures helps clarify which time points provide the strongest information about the benefit on one intervention condition over another.

All models were run in Mplus Version 4.1 (Muthén and Muthén, 1998–2006) with a minimum of 100 sets of start values. The number of starting values was increased if the best log likelihood was not replicated at least three times. In addition, a common finding in mixture analyses is the “label switching” of classes in the estimation process (Chung et al., 2004). Previous analyses had found that the three developmental classes differ in their starting point in fall of first grade, such as the *persistent high* class started at a higher intercept than the *escalating medium* class which started at a higher intercept than the *stable low* class (Muthén et al., 2002; Petras et al., 2004). We incorporated this knowledge in defining the inequality constraints so that label switching was avoided.

2.5. Missing data

The estimates of parameters in the models were adjusted for attrition. Mplus uses full information maximum likelihood estimation under the assumption that the data were missing at random. Missing at random assumes that the reason for the missing data is either random or random after incorporating other variables measured in the study (Arbuckle, 1996; Little, 1995). Full information maximum likelihood, used in the present study, is widely accepted as an appropriate way of handling missing data (Muthén and Shedden, 1999; Schafer and Graham, 2002). Overall, the percentage of the sample who had missing data at a given time point was similar for Cohorts 1 and 2 (see Table 1). Mplus bases its estimates on all available time points for a given case. To assess the extent of missing data in the dataset, Mplus provides a bivariate covariance “coverage” matrix that gives the proportion of available observations for each indicator variable and pairs of variables, respectively. The minimum coverage necessary for models to converge is 0.10 (Muthén and Muthén, 1998–2006). In the present study, coverage for males ranged from 0.437 to 0.925 in Cohort 1 and 0.381 to 0.923 in Cohort 2, and for females, 0.451 to 0.961 in Cohort 1 and 0.397 to 0.924 in Cohort 2, more than adequate for unbiased estimation.

3. Results

In this section, we present the results by cohort status separately by gender. In each instance we describe the intervention effect through seventh grade on the course of aggressive, disruptive behavior as well as the impact of the intervention into young adulthood on ASPD diagnosis and incidence of violent and criminal behavior.

3.1. Cohort 1 males

3.1.1. Model. A three-class solution fit the data the best, separating a *persistent high* class, an *escalating medium* class, and a *stable low* class based on aggressive, disruptive behavior from first through seventh grade. The *persistent high* class began with the highest rates of aggressive, disruptive behavior that rose through third or fourth grade, then decreased into middle school, but remained consistently higher throughout. The *escalating medium* class showed medium ratings in first grade and increasing aggressive, disruptive behavior through seventh grade which did not reach the high levels seen initially in the *persistent high* group. The *stable low* group began and continued with low ratings of aggressive, disruptive behavior. Both linear and quadratic slopes were needed to model the course of aggressive, disruptive behavior. The quadratic slope was treated as a fixed effect and intercept and slope were allowed to co-vary. The residual variances of the teacher ratings in first and second grade, when teachers rated students in both fall and spring each year, were free to correlate. The *stable low* group showed a smaller intercept and residual variance compared to the other classes.

3.1.2. Impact through seventh grade for Cohort 1 males. Fig. 1 shows that among the 199 Cohort 1 males, 14.3% belonged to the *persistent high* class, 50.2% to the *escalating medium* class, and 35.5% to the *stable low* class of aggressive, disruptive behavior (LL = −1603.831, # of parameters = 40). Among the *persistent high* group, GBG males displayed a lower slope compared to control males indicating that the GBG had an impact in lowering the growth of aggressive, disruptive behavior (β_{GBG} on slope = −0.440, $p = 0.041$). This positive GBG intervention effect was sustained until fourth grade with p -values ranging from 0.007 to 0.031. In fifth grade the mean difference between the two conditions was marginally significant (i.e., $p < 0.077$) and reached non-significance in sixth ($p < 0.315$) and seventh grade ($p < 0.615$). These significance values at each time point are based on univariate comparisons; when we corrected for the

multiple comparisons across time, we found that there remained significant benefits of the GBG at third and fourth grade.

We did not find a significant GBG impact in the other two classes; compared to control males, GBG males in the *escalating medium* and *stable low* classes had a lower slope of aggressive, disruptive behavior (*escalating medium* class: β_{GBG} on slope = −0.047, $p = 0.7106$; *stable low* class: β_{GBG} on slope = −0.002, $p = 0.9705$). In addition, GBG males compared to control males showed a non-significantly higher nonlinear slope. The effect reached from 0.053 ($p = 0.3281$) in the *persistent high* group to 0.006 ($p = 0.7505$) in the *escalating medium* and 0.001 ($p = 0.9116$) in the *stable low* class. Similar results were previously reported in Muthén et al. (2002).

3.1.3. Impact into young adulthood for Cohort 1 males. The young adult outcomes of ASPD and violent and criminal behavior were modeled as a function of class membership (see Fig. 2). In the *persistent high* class of control males all qualified for an ASPD diagnosis (LL = −1676.416, # of parameters = 46). Overall, males in the GBG condition showed a lower prevalence of ASPD across all three classes; however, the reduction in the prevalence was only significant in the *persistent high* class where 100% of the controls but only 40.1% of the GBG males had a diagnosis of ASPD. A likelihood ratio test confirmed this differences as being significant ($\chi^2 = 17.18$, d.f. = 1, $p < 0.001$). In the *escalating medium* class, GBG males were 15% less likely to be diagnosed with ASPD than control males (37.1% in GBG males, 41% in control males, OR = 0.851), and in the *stable low* class, GBG males were 32% less likely to be diagnosed with ASPD compared to the control males (17.8% in GBG males, 17.8% in control males, OR = 0.678). Neither of the latter two OR's were significantly different from one.

We found similar results for violent and criminal behavior (see Fig. 3). Thirty-four percent of GBG males in the *persistent high* class had both ASPD and a record of violent and criminal behavior and 9% had ASPD alone compared with 50% of

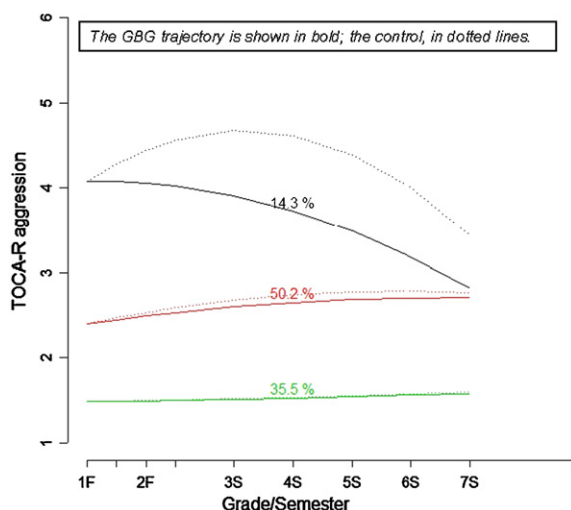


Fig. 1. Impact in Cohort 1 males through seventh grade ($N = 199$).

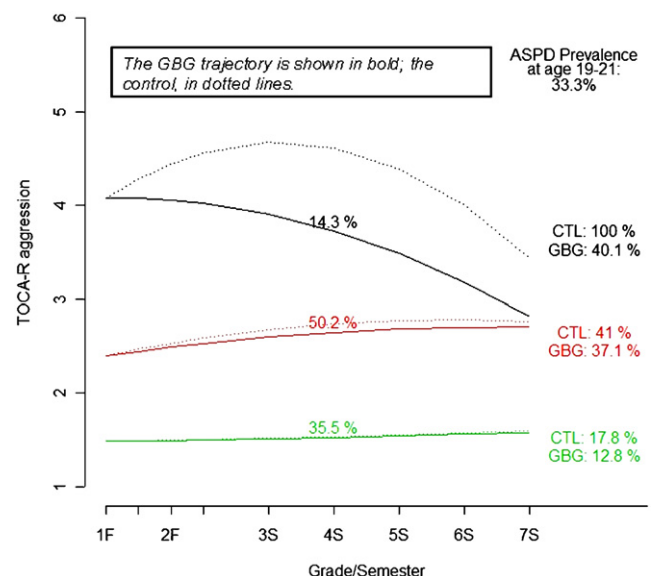


Fig. 2. Impact on ASPD in Cohort 1 males in young adulthood ($N = 199$).

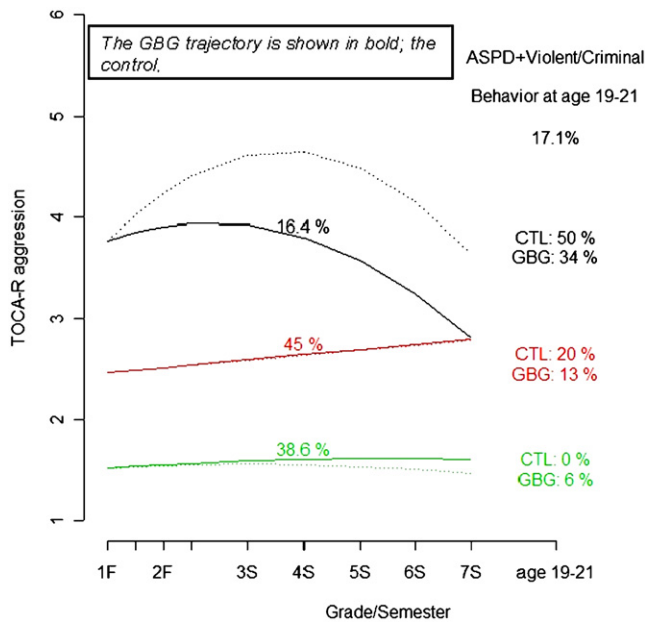


Fig. 3. Impact on ASPD and violent and criminal behavior combined in Cohort 1 males in young adulthood ($N = 199$).

persistent high control males who had both outcomes and 50% who had ASPD alone ($\chi^2 = 25.062$, d.f. = 3, $p < 0.001$). For the other less aggressive, disruptive classes the GBG impact was again not significant in reducing ASPD or violent and criminal behavior combined or individually.

3.2. Cohort 1 females

3.2.1. Model. The model set-up used for Cohort 1 females was very similar to the one used for Cohort 1 males, with the exception that the slope variance in the *stable low* class was non-significantly different from zero and was therefore set to zero. Importantly, the middle class for Cohort 1 females was not an escalating group as for Cohort 1 males, rather the medium ratings of aggressive, disruptive behavior persisted through seventh grade, lending the name of *persistent medium* to this class. It is also noteworthy that the intercept for the *persistent high* class was lower than that for the *persistent high* males.

3.2.2. Impact through seventh grade for Cohort 1 females. Fig. 4 shows the three-class solution that best fit the data ($LL = -1167.886$, # of parameters = 41). Of the 204 females, 8.4% belonged to the *persistent high* class, 55.4% belonged to a *persistent medium* class, and 36.3% were in the *stable low* class. GBG *persistent high* females showed a significantly lower slope than the *persistent high* control females, indicating a GBG impact in the hypothesized direction (β_{GBG} on slope = -0.498 , $p = 0.0356$). We further investigated the GBG impact among the *persistent high* females by a likelihood ratio test comparing the intervention's effect on the linear and quadratic growth combined ($\chi^2 = 5.439$, d.f. = 2, $p < 0.0659$) which yielded a marginally significant effect. As can be seen in Figure 4, the GBG impact for the *persistent high* females was not sustained through seventh grade. Unlike Cohort 1 males, by seventh grade

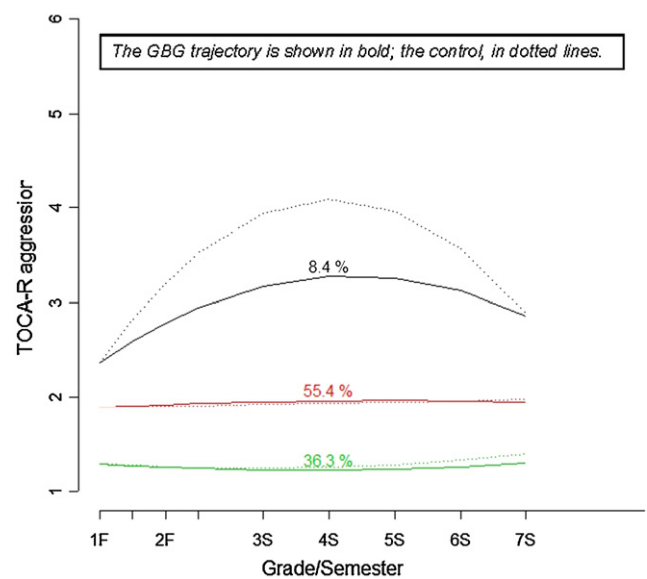


Fig. 4. Impact in Cohort 1 females through seventh grade ($N = 204$).

GBG females showed a similar level of aggressive, disruptive behavior when compared to controls. Based on these findings, we then examined whether for the *persistent high* females there was a significant difference in aggressive, disruptive behavior by intervention at each of the time points. We corrected for multiple comparisons and none of the differences at the different time points reached significance.

We did not find a significant GBG impact in the *persistent medium* ($p = 0.8173$) or the *stable low* classes ($p = 0.9585$). In addition, GBG females in the *persistent high* class showed a non-significantly higher nonlinear slope ($p = 0.0685$) and GBG females in the *persistent medium* ($p = 0.8220$) and *stable low* class ($p = 0.7879$) showed a non-significantly lower nonlinear slope.

3.2.3. Impact into young adulthood for Cohort 1 females. To explore the GBG impact among Cohort 1 females in young adulthood we used the same model set-up used for males (data not shown). This model ($LL = -1211.892$, # of parameters = 46) revealed a non-significant reduction in prevalence of ASPD in the *persistent high* and the *persistent medium* class. Among the *persistent high* females, 50.6% of the controls compared to 38.7% of the GBG group received a diagnosis for ASPD ($p = 0.7064$), and among the *persistent medium* females 14.1% of the controls compared to 4.2% of the GBG group had the outcome ($p = 0.1493$). In the *stable low* group 0% of the control females and 2.3% of the GBG females met criteria for the diagnosis of ASPD ($p = 0.3871$). To further explore this finding we conducted a series of likelihood ratio tests by comparing the current model to a model where the GBG impact was constrained to be equal across classes and to a model where the impact was set to zero. Ultimately, the zero impact model fit the data the best ($\chi^2 = 3.298$, d.f. = 3, $p = 0.3479$), indicating that the GBG intervention did not significantly reduce the risk for later ASPD for females compared to controls in any of the three classes.

Given the small prevalence of ASPD and reports of violent and criminal behavior among females and the smaller prevalence of *persistent high* aggressive, disruptive females in our sample, models using reports of violent and criminal behavior were very unstable and are therefore omitted from this manuscript. Among the Cohort 1 females, the majority qualified for neither ASPD nor violent and criminal behavior (86%). Of the remaining females, 2.3% had both outcomes, while 4.1% qualified for violent and criminal behavior only and 7.6% for ASPD only.

3.3. Cohort 2 males

3.3.1. Model. Two candidate models were identified as potentially appropriate for examining the GBG impact among Cohort 2 males. The first model used the same specifications as the best fitting model in Cohort 1, but since no assessment took place in fall of second grade the residual correlation between fall and spring of second grade was omitted. Additionally, the mean quadratic slope was set to zero in the *stable low* class. The best fitting model ($\chi^2 = 37.4$, d.f. = 1, $p < 0.001$) freed the residual variance in spring of first grade in the *escalating medium* class and was used for subsequent analysis. This model (LL = -1151.087, # of parameters = 39) yielded three trajectories similar to those of Cohort 1: a *persistent high* class which began with high aggressive, disruptive behavior that increased until third or fourth grade, then decreased through seventh grade yet still remained high; an *escalating medium* class that showed initial mid-level aggressive, disruptive behavior which increased through seventh grade; and a *stable low* class that exhibited low levels of aggressive, disruptive behavior from beginning of first grade through seventh grade.

3.3.2. Impact through seventh grade for Cohort 2 males. Among Cohort 2 males, 18% belonged to the *persistent high* class, 56.2% to the *escalating medium* class, and 25.8% to the *stable low* class (see Fig. 5). Overall the growth shape of these

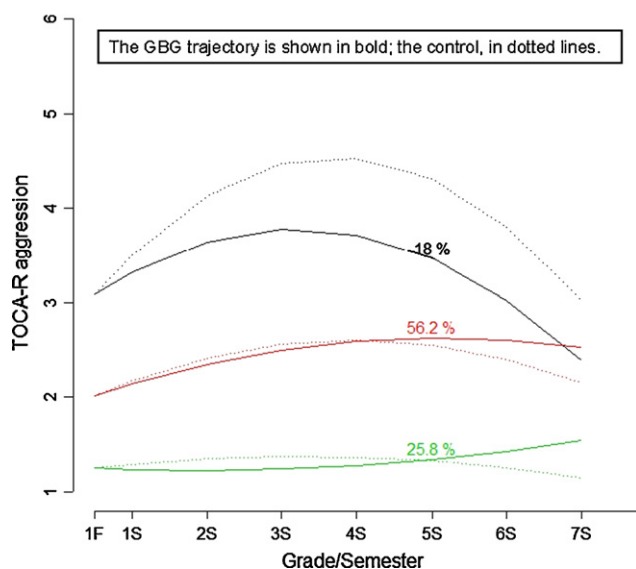


Fig. 5. Impact in Cohort 2 males through seventh grade ($N = 181$).

classes appeared quite similar to that of Cohort 1 while the prevalences are slightly higher for the *persistent high* and the *escalating medium* class and lower for the *stable low* class. In terms of the impact through seventh grade, we found significant overall lower growth of aggressive, disruptive behavior in the GBG *persistent high* class compared to controls ($\chi^2 = 6.024$, d.f. = 2, $p < 0.0492$ by likelihood ratio test comparing the intervention's effects on linear and quadratic growth combined). The *escalating medium* class showed a lower slope than the controls, but was not significantly different. In the *stable low* class we found a significant impact with GBG males showing a significantly lower slope compared to control males (β_{GBG} on slope = -0.122, $p = 0.0312$). Based on these finding of different growth trajectories for *persistent high* males in the GBG compared to control classrooms, we then examined whether there were significant differences in aggressive, disruptive behavior by intervention at each of the school grade levels in this high aggressive, disruptive class. As we did for Cohort 1, we corrected for multiple comparisons using the Bonferroni method. Unlike the results for Cohort 1, none of the differences at each different grade level reached the standard level of significance.

3.3.3. Impact into young adulthood for Cohort 2 males. As in Cohort 1, we examined impact on young adult outcomes for each class. This model (LL = -1234.017, # of parameters = 45) yielded a non-significant impact in all of the classes. Compared to their controls, GBG males in the *persistent high* class were slightly less likely to be diagnosed with ASPD (OR = 0.993) while GBG males in the *escalating medium* and *stable low* classes showed an increase in the prevalence of the diagnosis (OR = 1.102 and 3.138, respectively). To further investigate this finding, we tested the results against a model of zero impact (LL = -1234.606, # of parameters = 42) to find that the increase in ASPD among the GBG group in the *escalating medium* and *stable low* classes was not significantly different than no effect at all ($\chi^2 = 1.178$, d.f. = 3, $p = 0.7583$).

When testing the impact of the intervention on the outcomes of violent and criminal behavior and ASPD alone or in combination, we also did not find significant differences by intervention condition ($\chi^2 = 0.997$, d.f. = 3, $p = 0.8021$). However, the reductions in prevalence were in the hypothesized direction and the highest difference was found among males who qualified for ASPD and engaged in criminal and violent behavior.

3.4. Cohort 2 females

3.4.1. Model. The model set-up used for Cohort 2 females was identical to the one used for Cohort 2 males, except that the covariance between the intercept and slope was set to zero due to a very small and non-significant estimate. A three-trajectory solution fit the data the best (LL = -993.486, # of parameters = 37), consisting of a *persistent high* class which showed the highest levels of aggressive, disruptive behavior that subsequently decreased through seventh grade, a *persistent medium* group with medium aggressive, disruptive behavior, and a *stable low* class that had consistently low levels of aggressive, disruptive behavior from first grade through seventh grade. The impact

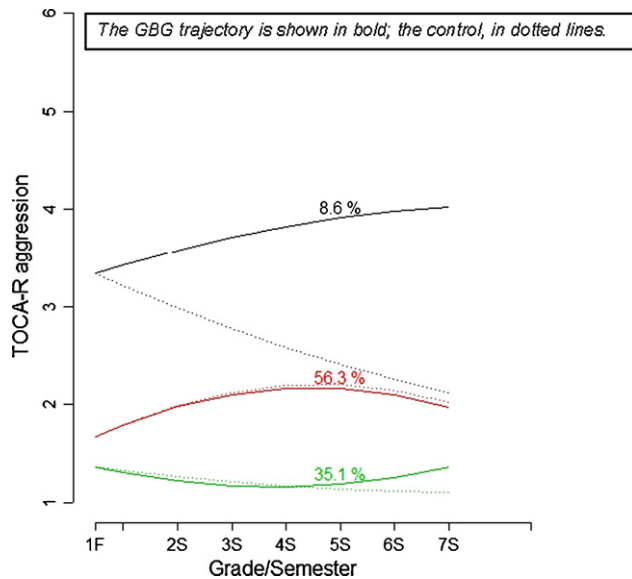


Fig. 6. Impact in Cohort 2 females through seventh grade ($N=184$).

on the nonlinear slope in the *persistent medium* class was also set to zero to avoid model instability.

3.4.2. Impact through seventh grade for Cohort 2 females.

Fig. 6 shows the three-class model for Cohort 2 females. A small *persistent high* group (8.6%) was distinguished from a *persistent medium* (56.3%) and a *stable low* class (35.1%). GBG females in the *persistent medium* and *stable low* classes showed a non-significantly lower slope in aggressive, disruptive behavior when compared to their controls (β_{GBG} on slope = -0.008 , $p=0.9936$; β_{GBG} on slope = -0.051 , $p=0.9593$, respectively). The GBG intervention had an opposite effect in the *persistent high* group (β_{GBG} on slope = 0.424 , $p=0.2793$). Further model testing revealed that a hypothesis of no GBG impact in any of the classes could not be rejected ($\chi^2=3.562$, d.f. = 2, $p=0.1715$).

3.4.3. Impact into young adulthood for Cohort 2 females.

Only 11.8% of the Cohort 2 females qualified for an ASPD diagnosis (LL = -1038.624 , # of parameters = 39) and all were in the *persistent medium* group. There was a non-significant increase in ASPD prevalence among the *persistent medium* GBG group; 16.4% of controls and 22.5% of GBG females received the diagnosis ($p=0.4959$). Further testing revealed that a hypothesis of zero impact compared to class unspecific impact could not be rejected ($\chi^2=0.653$, d.f. = 1, $p=0.4190$).

As stated before, given the small prevalence of ASPD and violent and criminal behavior among females and the overall smaller prevalence of *persistent high* aggressive, disruptive females in this sample, models using reports of violent and criminal behavior became very unstable and were therefore dropped. Among Cohort 2 females, 84.5% qualified for neither outcome. Of the 15.5% with an outcome, 2.7% had ASPD as well as violent and criminal behavior, 9.5% had ASPD only, and 3.4% had violent and criminal behavior only.

3.5. Summary of results

Cohort 1 GBG males in the *persistent high* trajectory showed significantly lower slopes of aggressive, disruptive behavior sustained overall through seventh grade and lower rates of ASPD and violence and criminal behavior by young adulthood compared to controls. These results show the GBG appeared to be effective in lowering early aggressive, disruptive behavior development and later prevalence of ASPD as well as ASPD combined with violent and criminal behavior among the *persistent high* aggressive, disruptive males. Positive GBG impact was also seen in males in the *escalating medium* and *stable low* classes; however, these results were not significant. *Persistent high* females showed a significantly lower slope of aggressive, disruptive behavior through seventh grade as well; however, the likelihood ratio test was only marginally significant and the mean differences between GBG and controls at each grade level were not significant. By young adulthood among females the reduction in ASPD prevalence did not reach significance although it was in the hypothesized direction. In the Cohort 2 replication, the GBG effect in the *persistent high* class of males on aggressive, disruptive behavior through seventh grade was significant and in the same direction as for Cohort 1. The effects were in the same direction by young adulthood but did not reach a significant reduction in ASPD and violent and criminal behavior alone or combined.

4. Discussion

This paper reports on the developmental epidemiological trajectories from first grade through early adolescence of early aggressive, disruptive classroom behavior and the strength of the trajectories in predicting antisocial personality disorder (ASPD) and violent and criminal behavior in young adulthood. We then report the impact of the GBG on each of the three developmental trajectories found to best typify the variation in courses leading to the young adulthood outcomes of ASPD and violent and criminal behavior. By following children from first grade through age 19–21, we investigated the effectiveness of using a universal preventive intervention like the GBG that is directed at early risk factors (e.g., aggressive, disruptive behavior) of later problem outcomes. This paper also is an important extension of the other work in this issue (Kellam et al., 2008) and on sixth grade conduct disorder (Brown et al., 2008) in two important ways. First, it investigates developmental trajectories of aggressive, disruptive behavior, not just initial first-grade ratings of aggressive, disruptive behavior. Second, the paper analyzes ASPD as well as violent and criminal behavior, two problem outcomes which represent a major public health and public safety concern. Importantly, violent and criminal behaviors were measured by using records of incarceration by young adulthood and therefore are going beyond self reports of the participating youth.

The GGMM analyses revealed three distinct trajectories for males, as hypothesized. There was a group that started high, rose through third and fourth grade, and then decreased somewhat by the end of seventh grade in aggressive, disruptive behav-

ior over time (i.e., the *persistent high* group). There emerged another group with initial moderate levels of aggressive, disruptive behavior that increased over elementary school but not to the higher level of the *persistent high* group (i.e., the *escalating medium* group). Finally, there was a group that started and remained at a low level of aggressive, disruptive behavior (i.e., the *stable low* group). Although with lower intercepts, similar trajectories were found for females in both cohorts, except the medium class did not display an increase in aggressive, disruptive behavior over time; rather it was a *persistent medium* group. Across the two cohorts the three trajectories were more similar than different, which lends additional support to the findings. Importantly, these growth trajectories were largely consistent with early starter and later starter models of antisocial behavior in keeping with the trajectories identified in the empirical literature (e.g., Broidy et al., 2003; Cote et al., 2001; Nagin and Tremblay, 1999; Petras et al., 2004; Schaeffer et al., 2003, 2006).

These population-based findings from an experimental intervention trial provide evidence consistent with the hypothesis that maladapting to classroom social task demands in the form of aggressive, disruptive behavior plays an etiological role in developing ASPD and violent and criminal behavior. The sources of these early social maladaptive responses to the classroom task demands can be genetic and/or learned and involve the teacher and whether she/he has the tools for adequate classroom behavior management and socialization of children to be students. A positive response to the GBG by the persistent higher risk males can lend important information as to the malleability of the causal processes, whether they may be in the child or the teachers' skills or both. In Cohort 2 little mentoring and monitoring of teacher's fidelity of GBG implementation was done, and by young adulthood the impact on ASPD as well as on violent and criminal behavior, while in the predicted direction did not reach significance. These findings strongly suggest the vulnerability of high risk children to poor classroom management by the teacher and the need for fidelity in delivering the GBG. The findings also reinforce the need for periodic and sufficiently long follow-up of preventive trials, covering major developmental periods and life transitions for a better understanding of developmental processes and prevention impact among males and females.

4.1. Reduction of impact in Cohort 2

The most likely explanation for the lack of significant GBG effects for Cohort 2 relative to Cohort 1 males is the decreased level of mentoring and monitoring in the second cohort (Kellam et al., 2008). While teachers in the effectiveness trial (i.e., Cohort 1) received 40 hours of training and support throughout the first year, during the following year those same teachers received no additional training and only minimal support for the replication/sustainability trial (i.e., Cohort 2). The reduced level of implementation is consistent with the beneficial but non-significant impact seen among the *persistent high* males in the second cohort by young adulthood.

Despite great similarities between the cohorts, it was also found that Cohort 2 males started at a lower level of first-grade aggressive, disruptive behavior compared to Cohort 1 males

($p < 0.010$). Cohort 2 males were more likely to come from economically disadvantaged families (as indicated by their higher likelihood to receive free lunch); therefore, it is possible that differences in the composition of the risk factors and sample distribution had a negative impact on the power to detect significant associations. Both hypotheses of lower level of implementation and higher individual/contextual risk require further study.

4.2. Gender differences in impact

Because the focus in this study was on overt and physical forms of aggressive, disruptive behavior, the lack of a GBG impact for females in young adulthood may be due to the overall lower rates of female aggressive, disruptive behavior and their lower prevalence in antisocial and violent outcomes. Indeed in our sample, the prevalence of females in the *persistent high* group was half of that found for males. In addition, in the *persistent high* class of Cohort 1, the intercept for females was lower than that of Cohort 1 *persistent high* males, indicating that females started at an overall lower level of aggressive, disruptive behavior as compared to males. However, as was shown by Schaeffer et al. (2006), the main difference is in the prevalence and not in the growth shape. Furthermore, the marginally significant impact among the *persistent high* females through seventh grade might indicate that highly aggressive, disruptive females respond to the universal intervention in a similar fashion as their male counterparts, lending some modest support to the life course/social field theory being applicable to both genders. It is quite possible, however, that the classroom levels of aggressive, disruptive behavior may impact males and females somewhat differently.

The lack of significant young adult impact among females is not surprising when we take into account findings from cross-sectional and short-term longitudinal studies of aggression and gender. This research has found that aggression in females is associated with peer rejection and depression during childhood (Crick, 1996; Crick and Grotpeter, 1995), but a link to serious antisocial behavior problems later in life has yet to be fully understood, especially in respect to the late starting group of females. In addition, it is possible there are gender differences in the early antecedent risk factors of the developmental courses that lead to later consequences. The course of aggressive, disruptive behavior may manifest itself differently in females, and it may not have the same salience during the course of development nor the same outcomes as found in males. With regards to the last possibility, researchers are starting to explore a broader array of deleterious long-term outcomes that might result from early female aggressive, disruptive behavior patterns, such as health problems (Serbin and Karp, 2004) and maladaptive parenting (Zoccolillo et al., 2004).

Our results for females might have been different if risks and outcomes more relevant to female development were included. For example, an assessment of relational aggression may have increased our understanding of female antisocial pathways. However, it has been found that relational and physical aggression tends to be highly correlated among both females and males (Henington et al., 1998; Tomada and Schneider, 1997), suggest-

ing that females who display high rates of physical aggression may also exhibit high rates of relational aggression. In fact, we found the course of aggressive, disruptive behavior among females differed more in terms of the prevalence than in the developmental pattern. Stronger results might also have been seen if the early antecedent measures included more parameters focused on internalizing behavior, such as depression, compared with more externalizing behavior, such as aggressive, disruptive behavior. Indeed, our previous work has shown that in females later psychological well-being (PWB) is most strongly linked to first-grade PWB (Kellam et al., 1983). Overall it is evident that there is a need for further research about early female aggression and developmental paths that could help more accurately explain the results seen in females with the GBG intervention.

4.3. *Strengths and limitations*

A major strength of this study is the design of the randomized intervention trial, which used an epidemiologically defined population of all youth entering first grade in 19 schools. The replication of the intervention with a second cohort adds strength to these analyses as well. Based on the design, we were able to effectively test hypotheses regarding the association between early aggressive, disruptive behavior and later ASPD and/or violent and criminal behavior as well as the impact of a preventive intervention on this association.

The current study focused on the associations of disruptive behaviors observed by teachers in the classroom, thus adding to research on the relationship between school attainment and social adjustment (Dishion, 1990). Although results from teacher ratings may not be equivalent to those from other sources, such as self, peer, or parent reports, our research as well as the work of other groups has shown that teacher ratings of aggressive, disruptive behavior have the highest level of agreement with student self reports and have equivalent levels of agreement to those found between parent and student self reports of problem behavior (Kellam et al., 1998; Lochman, 1995; Loeber et al., 1984).

Importantly, this research strongly supports predictions from life course/social field theory that early successful social adaptation in the face of prominent developmental challenges tends to promote later adaptation. In this paper we focused on a critical developmental challenge – the transition to the first-grade classroom with its classroom rules and required student behavior, and how this early transition leads to the course of development over elementary school and is related to later developmental outcomes such as ASPD and criminal and violent behavior. The results appear to support that under favorable intervention conditions (e.g., sufficient support and mentoring of teachers) males with early social maladaptive responses to school can be helped, and their risk for long-term acting-out negative outcomes subsequently can be reduced.

4.4. *Future research*

One direction for future research is to mount a trial where the GBG intervention is implemented in pre-school and/or other

grade levels in addition to first and second grades. Such extension might not only improve the impact among the *persistent high* group, but also reduce the risk for later aggressive, disruptive behavior and violent and criminal behavior among the *escalating medium* group. In addition, it is clearly important to study how to nest back-up services for those children who need more than the universal intervention based on children's responses or lack of responses to the universal. In this way, we could address the children who do not respond to the universal intervention and who need more.

We make the argument that the lack of a significant impact in the second cohort was due to lower levels of monitoring and mentoring. However, very little systematic information is available regarding sufficient levels of GBG implementation to detect an intervention impact. There is growing awareness that measuring quality of implementation is very necessary in preventive trials, particularly of universal interventions. The second generation trial in Baltimore demonstrated the vital role of the level of fidelity in implementing the intervention (Ialongo et al., 1999). Trials that randomize the level of implementation quality are one strategy, but costly in time and short in knowledge about what exactly to randomize. In the meantime, measuring the qualities and including variation in the implementation in the analyses of impact should now be considered requirements of rigorous evaluation and is a central focus in the third generation trial now proceeding in Baltimore.

In order to examine the generalizability of the findings presented in this study, replications involving populations varying in terms of ethnicity, socioeconomic status, and neighborhood or community characteristics will need to be conducted. We have suggested that the non-significant impact of the GBG among females into young adulthood in the *persisting high* class was likely due to lower prevalence of aggressive, disruptive behavior in females relative to males; therefore, larger samples are needed to provide for an adequately powered statistical test of the GBG impact on females. Finally, testing of the GBG impact on relational aggression may be informative, as would examining the GBG impact on relational young adult outcomes, such as partner violence or parenting. The third generation trial currently being conducted in Baltimore includes the GBG as well as measures of relational aggression.

Understanding variation in developmental processes within and across gender in epidemiologically defined populations is a cardinal requirement for the advancement of prevention programs. A major prevention strategy developed over the last several decades is based on uncovering early antecedents along developmental paths leading to successful and unsuccessful social adaptational trajectories, and then directing interventions at the antecedent mediators or moderators to promote successful adaptation and well-being to prevent problem outcomes (Kellam and Rebok, 1992; Kellam et al., 1999). The results reported here support this strategy, but they also point to missing areas of knowledge about development that would allow progress from this strategy to move ahead. Results reported here for males, particularly higher risk males, and the lesser impact for females call strongly for research to understand the shared and non shared developmental processes within and across genders.

Conflict of interest

The author and all of the co-authors declare that they have no conflict of interest.

Acknowledgements

We would like to extend our deepest gratitude to our partners, the Baltimore City Public School System, and the youths and families who participated in this project. We would also like to thank Drs. Sharon Lambert and Cindy Schaeffer for their insightful comments on earlier drafts of this manuscript as well as Dr. Kimberly Kendziora for her work with later drafts. We also thank Dr. Wei Wang for his assistance in testing the time specific impact of the intervention. In addition, we are grateful to the Prevention Science Methodology Group, led by Drs. C. Hendricks Brown and Bengt Muthén for their generous feedback on earlier analyses. Last, but not least we thank Amelia Mackenzie, who provided extensive and very insightful editorial and research assistance throughout the long process of checking and double checking data, references, and inferences, as well as for improving the readability of the text.

Role of funding source: During the last 22 years this research has been supported by NIMH and NIDA through the following grants: R01 MH42968, P50 MH38725, R01 MH40859, T32 MH018834, R01 DA09897, and R01 DA004392. The NIMH or NIDA had no further role in the analysis and interpretation of data or any conclusions.

Contributors: Dr. Petras led the statistical analyses, interpretation and the write-up of the findings as well as the preparation of the manuscript for submission as well as resubmission.

Dr. Kellam was P.I. of the grants that supported this work throughout the period of the trial and the young adult outcomes, with the exception of the Prevention Science Methodology grant led by Hendricks Brown. He led the Life Course/Social Field and Developmental Epidemiological conceptual frame that underlies the work, and led the trial reported here including the measures and choice of the intervention and its lead staff person and the precision of its implementation. He led the community and institutional base building and its maintenance including the core partnership with Baltimore City Public School System. He collaborated in defining the research question reported in this paper and the analytic strategies employed. Lastly, he participated in the writing of the manuscript and its final assessment prior to submission.

Drs. Brown and Muthén consulted with the first author regarding alternative model specification of the growth process of aggressive, disruptive behavior, and provided technical assistance in the application of the software package Mplus, as well as contributed to the interpretation and writing of results.

Dr. Ialongo assisted in the write-up through providing substantive edits and suggestions to this manuscript.

Dr. Poduska assisted in the write-up through providing substantive edits and suggestions to an earlier draft of the manuscript.

References

- American Psychiatric Association, 1994. Diagnostic and Statistical Manual of Mental Disorders, IV ed. American Psychiatric Association, Washington, DC.
- Arbuckle, J.L., 1996. Full information estimation in the presence of incomplete data. In: Marcoulides, G.A., Schumacker, R.E. (Eds.), *Advanced Structural Equation Modeling: Issues and Techniques*. Erlbaum, Mahwah, NJ.
- Baron, R.M., Kenny, D.A., 1986. The moderator-mediator variable distinction in social psychological research: conceptual, strategic and statistical considerations. *J. Pers. Soc. Psychol.* 51, 1173–1182.
- Barrish, H., Saunders, M., Wolf, M., 1969. Good behavior game: effects of individual contingencies for group consequences on disruptive behavior in a classroom. *J. Appl. Behav. Anal.* 2, 119–124.
- Bennett, K.J., Lipman, E.L., Brown, S., Racine, Y., Boyle, M.H., Offord, D.R., 1999. Predicting conduct problems—can high-risk children be identified in kindergarten and grade 1? *J. Consult. Clin. Psychol.* 4, 470–480.
- Bennett, K.J., Lipman, E.L., Racine, Y., Offord, D.R., 1998. Annotation: do measures of externalizing behavior in normal populations predict later outcome? Implications for targeted interventions to prevent conduct disorder. *J. Child Psychol. Psychiatry* 8, 1059–1070.
- Britt, C.L., 2000. Social context and racial disparities in punishment decisions. *Justice Quart.* 17, 707–732.
- Broidy, L.M., Nagin, D.S., Tremblay, R.E., Bates, J.E., Brame, B., Dodge, K.A., Fergusson, D., Horwood, J.L., Loeber, R., Laird, R., Lyman, D.R., Moffitt, T.E., Pettit, G.S., Vitaro, F., 2003. Developmental trajectories of childhood disruptive behaviors and adolescent delinquency: a six-site, cross-national study. *Dev. Psychol.* 39, 222–245.
- Brook, J.S., Nomura, C., Cohen, P., 1989. A network of influences on adolescent drug involvement: neighborhood, school, peer, and family. *Genet. Soc. Gen. Psychol. Monogr.* 115, 125–145.
- Brown, C.H., 1993. Statistical methods for preventive trials in mental health. *Stat. Med.* 12, 289–300.
- Brown, C.H., Liao, J., 1999. Principles for designing randomized preventive trials in mental health: an emerging developmental epidemiology paradigm. *Am. J. Commun. Psychol.* 27, 673–710.
- Brown, C.H., Wang, W., Kellam, S.G., Muthén, B.O., Petras, H., Toyinbo, P., Poduska, J., Ialongo, N., Wyman, P.A., Chamberlain, P., Sloboda, Z., MacKinnon, D., Windham, A., The Prevention Science and Methodology Group, 2008. Methods for testing theory and evaluating impact in randomized field trials: intent-to-treat analyses for integrating the perspectives of person, place, and time. *Drug Alcohol Depend.* 95, S74–S104.
- Capaldi, D.M., Stoolmiller, M., 1999. Co-occurrence of conduct problems and depressive symptoms in early adolescent boys. III. Prediction to young adult adjustment. *Dev. Psychopathol.* 11, 59–84.
- Chaiken, J.M., Chaiken, M.R., 1990. Drugs and predatory crime. In: Tonry, M., Wilson, J.Q. (Eds.), *Crime and Justice: A Review of Research*, vol. 13: *Drugs and Crime*. University of Chicago Press, Chicago, pp. 203–239.
- Chung, H., Loken, E., Schafer, J.L., 2004. Difficulties in drawing inferences with finite-mixture models: a simple example with a simple solution. *Am. Statist.* 2, 152–158.
- Cicchetti, D., Schneider-Rosen, K., 1984. Toward a transactional model of childhood depression. In: Cicchetti, D., Schneider-Rosen, K. (Eds.), *Childhood Depression: A Developmental Perspective*. Jossey-Bass, San Francisco, pp. 5–28.
- Cohen, M.A., Miller, T.R., 1998. The cost of mental health care for victims of crime. *J. Interpers. Violence* 13, 93–110.
- Compton, W.M., Conway, K.P., Stinson, F.S., Colliver, J.D., Grant, B.F., 2005. Prevalence, correlates, and comorbidity of DSM-IV antisocial personality syndromes and alcohol and specific drug use disorders in the United States: results from the National Epidemiologic Survey on alcohol and related conditions. *J. Clin. Psychiatry* 66, 677–685.
- Cote, S., Zoccolillo, M., Tremblay, R.E., Nagin, D., Vitaro, F., 2001. Predicting girls' conduct disorder in adolescence from childhood trajectories of disruptive behaviors. *J. Am. Acad. Child Adolesc. Psychiatry* 40, 678–684.
- Crick, N.R., 1996. The role of overt aggression, relational aggression, and prosocial behavior in the prediction of children's future social adjustment. *Child Dev.* 67, 2317–2327.

- Crick, N.R., Grotpeter, J.K., 1995. Relational aggression, gender, and social-psychological adjustment. *Child Dev.* 66, 710–722.
- Dishion, T.J., 1990. The family ecology of boys' peer relations in middle childhood. *Child Dev.* 61, 874–892.
- Dishion, T.J., Spracklen, K.M., Andrews, D.W., Patterson, G.R., 1996. Deviancy training in male adolescent friendships. *Behav. Ther.* 27, 373–390.
- Dodge, D., 1999. Early prevention and intervention equals delinquency prevention. *Focal Point* 13, 17–21.
- Dolan, L.J., Kellam, S.G., Brown, C.H., Werthamer-Larsson, L., Rebok, G.W., Mayer, L.S., Laudolff, J., Turkkan, J., Ford, C., Wheeler, L., 1993. The short-term impact of two classroom-based preventive interventions on aggressive and shy behaviors and poor achievement. *J. Appl. Dev. Psychol.* 14, 317–345.
- Eddy, J.M., Reid, J.B., Fetrow, R.A., 2000. An elementary school-based prevention program targeting modifiable antecedents of youth delinquency and violence: Linking the Interests of Families and Teachers (LIFT). *J. Emot. Behav. Disord.* 8, 165–176.
- Ensminger, M.E., Kellam, S.G., Rubin, B.R., 1983. School and family origins of delinquency: comparisons by sex. In: Van Dusen, K.T., Mednick, S.A. (Eds.), *Prospective Studies of Crime and Delinquency*. Kluwer-Nijhoff Publishing, Boston, pp. 73–97.
- Ensminger, M.E., Slusarcick, A.L., 1992. Paths to high school graduation or dropout: a longitudinal study of first grade cohort. *Sociol. Educ.* 65, 95–113.
- Farrington, D.P., 1995. The development of offending and antisocial behavior from childhood: key findings from the Cambridge Study in Delinquent Development. *J. Child Psychol. Psychiatry* 36, 929–964.
- Farrington, D.P., Gunn, J. (Eds.), 1985. *Aggression and Dangerousness*. John Wiley & Sons, New York.
- Farrington, D.P., Welsh, B.C., 2005. Randomized experiments in criminology: what have we learned in the last two decades? *J. Exp. Criminol.* 1, 9–38.
- Fazel, S., Danesh, J., 2002. Serious mental disorder in 23000 prisoners: a systematic review of 62 surveys. *Lancet* 359, 545–550.
- Gottfredson, D.C., 2001. *Schools and Delinquency*. Cambridge University Press, Cambridge, United Kingdom.
- Gottfredson, D.C., Wilson, D.B., Najaka, S.S., 2002. School-based crime prevention. In: Sherman, L.W., Farrington, D.P., Welsh, B.C., MacKenzie, D.L. (Eds.), *Evidence-based Crime Prevention*. Routledge, London.
- Greenberg, P., Kusche, C., 1998. Preventive intervention for school-age deaf children: the PATHS curriculum. *J. Deaf. Stud. Deaf Educ.* 3, 49–63.
- Grossman, D.C., Neckerman, H.J., Koepsell, T.D., Liu, P.Y., Asher, K.N., Beland, K., Frey, K., Rivara, F.P., 1997. Effectiveness of a violence prevention curriculum among children in elementary school: a randomized controlled trial. *JAMA* 277, 1605–1611.
- Hawkins, J.D., Guo, J., Hill, K.G., Battin-Pearson, S., 2000a. Long term effects of the Seattle Social Development Intervention on school bonding trajectories. In: Maggs, J., Schulenberg, J. (Eds.), *Applied Developmental Science: Special Issue: Prevention as Altering the Course of Development* 5, pp. 225–236.
- Hawkins, J.D., Herrenkohl, T.I., Farrington, D.P., Brewer, D., Catalano, R.F., Harachi, T.W., Cothorn, L., 2000b. Predictors of youth violence. *Juvenile Justice Bulletin*, Office of Juvenile Justice and Delinquency Prevention, 1–11.
- Hawkins, J.D., Weiss, J.G., 1985. The social development model: an integrated approach to delinquency prevention. *J. Prim. Prev.* 6, 73–97.
- Henington, C., Hughes, J.N., Cavell, T.A., Thompson, B., 1998. The role of relational aggression in identifying aggressive boys and girls. *J. School Psychol.* 36, 457–477.
- Ialongo, N.S., Werthamer, L., Kellam, S.G., Brown, C.H., Wang, S., Lin, Y., 1999. Proximal impact of two first-grade preventive interventions on the early risk behaviors for later substance abuse, depression, and antisocial behavior. *Am. J. Community Psychol.* 27, 599–641.
- Jessor, R., Jessor, S.L., 1978. Theory testing on longitudinal research. In: Kandel, D.B. (Ed.), *Longitudinal Research in Drug Use: Empirical Findings and Methodological Issues*. Hemisphere Publishing Corp., Washington, DC.
- Keenan, K., Shaw, D., 1997. Developmental and social influences on young girls' early problem behavior. *Psychol. Bull.* 121, 95–113.
- Kellam, S.G., Anthony, J.C., 1998. Targeting early antecedents to prevent tobacco smoking: findings from an epidemiologically based randomized field trial. *Am. J. Public Health* 88, 1490–1495.
- Kellam, S.G., Branch, J.D., Agrawal, K.C., Ensminger, M.E., 1975. *Mental Health and Going to School: The Woodlawn Program of Assessment, Early Intervention, and Evaluation*. University of Chicago Press, Chicago.
- Kellam, S.G., Brown, C.H., Poduska, J., Ialongo, N., Wang, W., Toyinbo, P., Petras, H., Ford, C., Windham, A., Wilcox, H.C., 2008. Effects of a universal classroom behavior management program in first and second grades on young adult behavioral, psychiatric, and social outcomes. *Drug Alcohol Depend.* 95, S5–S28.
- Kellam, S.G., Brown, C.H., Rubin, B.R., Ensminger, M.E., 1983. Paths leading to teenage psychiatric symptoms and substance abuse: developmental epidemiological studies in Woodlawn. In: Guze, S.B., Earls, F.J., Barrett, J.E. (Eds.), *Childhood Psychopathology and Development*. Raven Press, New York, pp. 17–51.
- Kellam, S.G., Koretz, D., Moscicki, E.K., 1999. Core elements of developmental epidemiologically based prevention research. *Am. J. Commun. Psychol.* 27, 463–482.
- Kellam, S.G., Langevin, D.J., 2003. A framework for understanding "evidence" in prevention research and programs. *Prev. Sci.* 4, 137–153.
- Kellam, S.G., Ling, X., Merisca, R., Brown, C.H., Ialongo, N., 1998. The effect of the level of aggression in the first grade classroom on the course and malleability of aggressive behavior into middle school. *Dev. Psychopathol.* 10, 165–185 (see also Kellam, S.G., Ling, X., Merisca, R., Brown, C.H., Ialongo, N., 2000. The effect of the level of aggression in the first grade classroom on the course and malleability of aggressive behavior into middle school: results of a developmental epidemiology-based prevention trial: erratum. *Dev. Psychopathol.* 12, 107.).
- Kellam, S., Rebok, G., 1992. Building developmental and etiological theory through epidemiological based preventive intervention trials. In: McCord, J., Tremblay, R.E. (Eds.), *Preventing Antisocial Behavior: Interventions from Birth through Adolescence*. Neale Watson Academic Publishers, New York, pp. 162–195.
- Kellam, S.G., Rebok, G.W., Ialongo, N., Mayer, L.S., 1994. The course and malleability of aggressive behavior from early first grade into middle school: results of a developmental epidemiologically-based preventive trial. *J. Child Psychol. Psychiatry* 35, 259–282.
- Kellam, S.G., Werthamer-Larsson, L., Dolan, L.J., Brown, C.H., Mayer, L.S., Rebok, G.W., Anthony, J.C., Laudolff, J., Edelsohn, G., Wheeler, L., 1991. Developmental epidemiologically-based preventive trials: baseline modeling of early target behaviors and depressive symptoms. *Am. J. Commun. Psychol.* 19, 563–584.
- Kershaw, T., 1992. The effects of educational tracking on the social mobility of African Americans. *J. Black Stud.* 23, 152–169.
- Kessler, R.C., McGonagle, K.A., Zhao, S., Nelson, C.B., Hughes, M., Eshleman, S., Wittchen, H.U., Kendler, K.S., 1994. Lifetime and 12-month prevalence of DSM-III-R psychiatric disorders in the United States: results from the National Comorbidity Survey. *Arch. Gen. Psychiatry* 51, 8–19.
- Krueger, R.F., Caspi, A., Moffitt, T.E., Silva, P.A., 1998. The structure and stability of common mental disorders (DSM-III-R): a longitudinal-epidemiological study. *J. Abnorm. Child Psychol.* 107, 216–227.
- Lambert, E.W., Wahler, R.G., Andrade, A.R., Bickman, L., 2001. Looking for the disorder in conduct disorder. *J. Abnorm. Psychol.* 110, 110–123.
- Little, R.J., 1995. Modeling the dropout mechanism in repeated-measures studies. *J. Am. Stat. Assoc.* 90, 1112–1121.
- Lochman, J.E., The Conduct Problems Prevention Research Group, 1995. Screening of child behavior problems for prevention programs at school entry. *J. Consult. Clin. Psychol.* 63, 549–559.
- Loeber, R., Burke, J.D., Lahey, B.B., Winters, A., Zera, M., 2000. Oppositional defiant and conduct disorder: a review of the past 10 years. Part I. *J. Am. Acad. Child Adolesc. Psychiatry* 39, 1468–1484.
- Loeber, R., Dishion, T., Patterson, G.R., 1984. Multiple gating: a multistage assessment procedure for identifying youths at risk for delinquency. *J. Res. Crime Delinquency* 21, 7–32.
- Loeber, R., Farrington, D.P., Stouthamer-Loeber, M., Moffitt, T.E., Caspi, A., 1998. The development of male offending: key findings from the first decade of the Pittsburgh Youth Study. *Stud. Crime Crime Prev.* 7, 141–171.
- Loeber, R., Stouthamer-Loeber, M., 1998. Development of juvenile aggression and violence. Some common misconceptions and controversies. *Am. Psychol.* 53, 242–259.

- Magnusson, D., 1998. The logic and implications of a person-oriented approach. In: Cairns, R., Bergman, L., Kagan, J. (Eds.), *Methods and Models for Studying the Individual*. Sage Publishing, Thousand Oaks, CA, pp. 3–64.
- McCord, J., Ensminger, M.E., 1997. Multiple risks and co-morbidity in an African-American population. *Crim. Behav. Mental Health* 7, 339–352.
- Moffitt, T.E., 1993. Adolescence-limited and life-course-persistent antisocial behavior: a developmental taxonomy. *Psychol. Rev.* 100, 674–701.
- Moffitt, T.E., Caspi, A., Dickson, N., Silva, P., Stanton, W., 1996. Childhood-onset versus adolescent-onset antisocial conduct problems in males: natural history from ages 3 to 18 years. *Dev. Psychopathol.* 8, 399–424.
- Moffitt, T.E., Caspi, A., Harrington, H., Milne, B.J., 2002. Males on the life-course-persistent and adolescence-limited antisocial pathways: follow-up at age 26 years. *Dev. Psychopathol.* 14, 179–207.
- Moran, P., 1999. The epidemiology of antisocial personality disorder. *Soc. Psychiatry Psychiatr. Epidemiol.* 34, 231–242.
- Mrazek, P.B., Haggerty, R.J., 1994. *Reducing Risks for Mental Disorders: Frontiers for Preventive Intervention Research*. National Academy Press, Washington, DC.
- Muthén, B., 2004. Latent variable analysis: growth mixture modeling and related techniques for longitudinal data. In: Kaplan, D. (Ed.), *Handbook of Quantitative Methodology for the Social Sciences*. Sage Publications, Newbury Park, CA, pp. 345–368.
- Muthén, B., Muthén, L.K., 1998–2006. *Mplus Users Guide*. Muthén and Muthén, Los Angeles.
- Muthén, B., Shedden, K., 1999. Finite mixture modeling with mixture outcomes using the EM algorithm. *Biometrics* 6, 463–469.
- Muthén, B.O., Brown, C.H., Masyn, K., Jo, B., Khoo, S.T., Yang, C.C., Wang, C.P., Kellam, S.G., Carlin, J.B., 2002. General growth mixture modeling for randomized preventive interventions. *Biostatistics* 3, 459–475.
- Nagin, D., Tremblay, R.E., 1999. Trajectories of boys' physical aggression, opposition, and hyperactivity on the path to physically violent and nonviolent juvenile delinquency. *Child Dev.* 70, 1181–1196.
- Nagin, D.S., Farrington, D.P., Moffitt, T.E., 1995. Life-course trajectories of different types of offenders. *Criminology* 33, 111–139.
- New, M., Berliner, L., 2000. Mental health service utilization by victims of crime. *J. Trauma. Stress* 13, 693–708.
- Offord, D.R., Bennett, K.J., 1997. Conduct disorder: long-term outcomes and intervention effectiveness. *J. Am. Acad. Child Adolesc. Psychiatry* 33, 1069–1078.
- Olweus, D., 1991. Bully/victim problems among school children: some basic facts and effects of a school based intervention program. In: Pepler, D., Rubin, K. (Eds.), *The Development and Treatment of Childhood Aggression*. Erlbaum, Hillsdale, NJ, pp. 411–438.
- Olweus, D., 1992. Victimization by peers: antecedents and long term outcomes. In: Rubin, K.H., Asendorf, J.B. (Eds.), *Social Withdrawal, Inhibition and Shyness in Children*. Erlbaum, Hillsdale, NJ, p. 17.
- Olweus, D., Alsaker, F.D., 1991. Assessing change in a cohort longitudinal study with hierarchical data. In: Magnusson, D., Bergman, L., Rudinger, G., Torestad, B. (Eds.), *Problems and Methods in Longitudinal Research*. Cambridge University Press, New York.
- Patterson, G.R., 1982. *Coercive Family Process*. Castalia, Eugene, OR.
- Patterson, G.R., Forgatch, M.S., Voerger, K.L., Stoolmiller, M., 1998. Variables that initiate and maintain an early-onset trajectory of offending. *Dev. Psychopathol.* 10, 531–547.
- Patterson, G.R., Reid, J.B., Dishion, T.J., 1992. *Antisocial Boys*. Castalia, Eugene, OR.
- Petras, H., Schaeffer, C., Ialongo, S., Muthén, B., Lambert, S., Poduska, J., Kellam, S., 2004. When the course of aggressive behavior in childhood does not predict antisocial outcomes in adolescence and adulthood: an examination of potential explanatory variables. *Dev. Psychopathol.* 16, 919–941.
- Poduska, J., Kellam, S., Wang, W., Brown, C.H., Ialongo, N., Toyinbo, P., 2008. Impact of the good behavior game, a universal classroom-based behavior intervention, on young adult service use for problems with emotions, behavior, or drugs or alcohol. *Drug Alcohol Depend.* 95, S29–S44.
- Rebok, G.W., Hawkins, W.E., Krener, P., Mayer, L.S., Kellam, S.G., 1996. Effect of concentration problems on the malleability of children's aggressive and shy behaviors. *J. Am. Acad. Child Adolesc. Psychiatry* 35, 193–203.
- Reid, J.B., Eddy, J.M., Fetrow, R.A., Stoolmiller, M., 1999. Description and immediate impacts of a preventive intervention for conduct problems. *Am. J. Commun. Psychol.* 27, 483–517.
- Reid, W.H., Thorne, S.A., 2006. Treating antisocial syndromes. *J. Psychiatr. Pract.* 12, 320–323.
- Reiss, D., Price, R.H., 1996. National research agenda for prevention research. The National Institute for Mental Health Report. *Am. Psychol.* 51, 1109–1115.
- Robins, L.N., 1966. *Deviant Children Grown Up*. Wilkens & Wilkens, Baltimore.
- Robins, L.N., 1978. Sturdy childhood predictors of adult antisocial behavior: replications from longitudinal studies. *Psychol. Med.* 8, 611–622.
- Robinson, F., Keithley, J., 2000. The impacts of crime on health and health services: a literature review. *Health Risk Soc.* 2, 253–266.
- Sameroff, A., 1994. Developmental systems and family functioning. In: Parke, R.D., Kellam, S.G. (Eds.), *Exploring Family Relationships with Other Social Contexts*, vol. 8. Lawrence J. Erlbaum Associates, Hillsdale, NJ, pp. 199–214.
- Satcher, D., 2001. *Youth Violence: A Report of the Surgeon General*. U.S. Department of Health and Human Services, Washington, DC.
- Schaeffer, C.M., Petras, H., Ialongo, N., Masyn, K.E., Hubbard, S., Poduska, J., Kellam, S., 2006. A comparison of girls' and boys' aggressive-disruptive behavior trajectories across elementary school: prediction to young adult antisocial outcomes. *J. Consult. Clin. Psychol.* 3, 500–510.
- Schaeffer, C.M., Petras, H., Ialongo, N., Poduska, J., Kellam, S., 2003. Modeling growth in boys aggressive behavior across elementary school: links to later criminal involvement, conduct disorder, and antisocial personality disorder. *Dev. Psychol.* 39, 1020–1035.
- Schafer, J.L., Graham, J.W., 2002. Missing data: our view of the state of the art. *Psychol. Methods* 7, 147–177.
- Schwartzman, A.E., Ledingham, J.E., Serbin, L.A., 1985. Identification of children at-risk for adult schizophrenia: a longitudinal study. *Int. Rev. Appl. Psychol.* 34, 363–380.
- Serbin, L.A., Karp, J., 2004. The intergenerational transfer of psychosocial risk: mediators of vulnerability and resilience. *Annu. Rev. Psychol.* 55, 333–363.
- Silverthorn, P., Frick, P.J., 1999. Developmental pathways to antisocial behavior: the delayed-onset pathway in girls. *Dev. Psychopathol.* 11, 101–126.
- Smith, P.K., Sharpe, S., 1994. The problem of school bullying. In: Smith, P.K., Sharpe, S. (Eds.), *School Bullying*. Routledge, London.
- Tomada, G., Schneider, B.H., 1997. Relational aggression, gender, and peer acceptance: invariance across culture, stability over time, and concordance among informants. *Dev. Psychol.* 33, 601–609.
- Turner, R.J., Gil, A., 2002. Psychiatric and substance use disorders in south Florida. *Arch. Gen. Psychiatry* 59, 43–50.
- Webster-Stratton, C., Taylor, T., 2001. Nipping early risk factors in the bud: preventing substance abuse, delinquency, and violence in adolescence through interventions targeted at young children (0–8 years). *Prev. Sci.* 2, 165–192.
- Werthamer-Larsson, L., Kellam, S.G., Wheeler, L., 1991. Effect of first grade classroom environment on child shy behavior, aggressive behavior, and concentration problems. *Am. J. Commun. Psychol.* 19, 585–602.
- Westermeyer, J., Thuras, P., 2005. Association of antisocial personality disorder and substance disorder morbidity in a clinical sample. *Am. J. Drug Alcohol Abuse* 31, 93–110.
- Wilcox, H.C., Kellam, S.G., Brown, C.H., Poduska, J., Ialongo, N.S., Wang, W., Anthony, J., 2008. The impact of two universal randomized first- and second-grade classroom interventions on young adult suicide ideation and attempts. *Drug Alcohol Depend.* 95, S60–S73.
- Zoccolillo, M., Paquette, D., Azar, R., Cote, S., Tremblay, R., 2004. Parenting as an important outcome of conduct disorder in girls. In: Putallaz, M., Bierman, K.L. (Eds.), *Aggression, Antisocial Behavior, and Violence among Girls*. Guilford, New York, pp. 242–261.